

Discussion 5: Virtual Memory Case Studies

Discussion Topic:

In your initial post, explain the benefits and downsides in using virtual memory in the OS.

- How does the rapidly increasing hardware capacity affect these tradeoffs?
 - Is there more or less necessity to use virtual memory in present-day systems than 20 or 30 years ago?
- Present 3-4 case studies to back up your assertions on this matter.

My Post:

Hello Class,

Virtual memory is an Operating System (OS) memory management system that provides the appearance of a large, continuous block of memory to applications, even if the primary memory (RAM) is limited and not provided in a contiguous manner (GeeksforGeeks, 2026). In other words, virtual memory is not just a way to “add more RAM” to a computer system; it is a major part of an OS’s memory management. Virtual memory uses both RAM (primary memory) and secondary memory, such as hard drives or SSDs, to allow large programs or multiple programs to be run concurrently when the primary memory is not enough to hold everything at once (CSU Global, n.d.).

Virtual memory is extremely flexible; this flexibility is the major benefit of implementing virtual memory within an OS memory management system. It allows the OS to run multiple applications concurrently more efficiently without requiring every part of every process to remain in RAM at the same time, as the processes are given their own virtual address space and do not have to be terminated to free RAM memory. Virtual memory systems use demand paging, which is a method that keeps pages in secondary memory until they are needed and then brings them into main memory when a reference is made to those pages (CSU Global, n.d.). Additionally, virtual memory is often used with a shared virtual memory system, where multiple programs can share the same common code or data instead of loading separate copies of it into memory. This combination makes an OS memory management system significantly more efficient as it improves the OS multitasking capabilities and overall use of the primary memory.

However, virtual memory has a downside; if an OS memory management system over-dependes on it, it can hurt the overall performance of the system and prematurely age the secondary memory component, especially SSDs, because of additional write activity. At that point, it is probably time to upgrade the system RAM capacity. In other words, when a program needs a page that is not currently in memory, a page fault occurs, and the OS must bring that page into RAM. This adds overhead and can slow the system down (CSU Global, n.d.). If too many pages are constantly moved between RAM and storage, the system can spend more time switching pages than actually processing program instructions, and this constant read/write activity can prematurely age the secondary memory component used as storage. Updating the RAM and storage capacity of a system usually helps with it; however, it does not completely eliminate it.

Compared to 20 or 30 years ago, computer systems today have much more RAM and faster storage. For example, SSDs are much faster than traditional hard drives. As a result, the slow processing of programs caused by virtual memory overhead is less noticeable in modern systems than it was on older systems. However, modern software applications also use much more memory because of this increased hardware capacity and because users expect software to do more at the same time. For example, web browsers,

games, virtual machines, data science tools, and AI workloads can put heavy pressure on memory systems, even more than older applications did. In other words, virtual memory was first implemented as an emergency memory substitute for RAM, but today it is implemented as a necessary and normal memory component of OS memory management systems, rather than only an emergency memory backup.

For example:

Windows implements a page file system that helps support system-committed virtual memory and can also support crash dumps (Microsoft, 2026). This means that even a modern Windows OS platform with a large amount of RAM may still rely on virtual memory. To handle peak memory pressure and system commit limits, but also to support reliability and system recovery. In this case, the page file is not only a backup for when RAM runs low; it is also part of how Windows keeps the system stable when applications request more committed memory than physical RAM alone can safely support.

Android memory management system uses a combination of RAM, zRAM, and storage to handle low-memory situations. However, the management system does not use storage for memory swap in the same way as some Linux systems; Windows does not implement zRAM. Android systems are usually memory-constrained systems needing to avoid frequent writes that can wear out storage (Android Developers, n.d.). Instead, they implement a zRAM system that compresses memory pages in RAM itself. This is interesting because the system still uses virtual memory but adds an extra layer to reduce memory pressure without relying heavily on storage switching. This shows how memory management and virtual memory have evolved and adapted to new hardware with memory constraints. The main difference between the Linux kernel zRAM and Android zRAM is not the technology itself per se, but how it is used. Linux zRAM is a general kernel feature that can be easily configured to match workflows (The Linux Kernel documentation. n.d.). Android zRAM, although it is basically the same basic kernel feature, is optimized for mobile devices and compresses swap app/process in RAM rather than relying heavily on storage-based compressed memory swap.

Virtual machines are now part of increasingly IT infrastructure, especially in a cloud computing environment, and more recently with increasing use of AI agents such as OpenClaw and Hermes Agent. Rupabhinda (2018) explains that virtualization is useful for resource consolidation because it allows multiple virtual machines to run on the same physical host system, but the more virtual machines are added, the more memory becomes scarce, which means virtualization needs careful memory management. Techniques to improve memory utilization can be implemented, such as ballooning, page sharing, memory overcommitment, and hot plug, which can help reclaim or reallocate memory between virtual machines. However, these techniques can affect performance negatively as their effectiveness heavily depends on application workflows.

Data-intensive workloads such as machine learning, data analytics, and large-scale database processing can place significant pressure on virtual memory systems. Kanellopoulos et al. (2025) explain that virtual memory overhead due to translation and memory allocation can take up a large part of the application execution time in some data-intensive workloads. Additionally, for such applications, better or more powerful hardware does not automatically translate to better performance. As these applications and datasets grow over time, they often have uneven workloads and workload time spikes that create significant memory fragmentation in most systems.

-Alex

References:

Android Developers. (n.d.). *Memory allocation among processes*. Google for Developers. <https://developer.android.com/topic/performance/memory-management>

CSU Global. (n.d.). *Module 5: Virtual memory* [Interactive Lecture]. CSC 507 Foundations of Operating Systems. Canvas.

GeeksforGeeks. (2026, April 15). *Virtual memory in operating system*. GeeksforGeeks. <https://www.geeksforgeeks.org/operating-systems/virtual-memory-in-operating-system/>

Kanellopoulos, K., Sgouras, K., Bostanci, F. N., Kakolyris, A. K., Konar, B. K., Bera, R., Sadrosadati, M., Kumar, R., Vijaykumar, N., & Mutlu, O. (2025). Virtuoso: Enabling fast and accurate virtual memory research via an imitation-based operating system simulation methodology. *Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2*. <https://doi.org/10.1145/3676641.3716027>

Microsoft. (2026, February 12). *Introduction to page files*. Microsoft Learn. <https://learn.microsoft.com/en-us/troubleshoot/windows-client/performance/introduction-to-the-page-file>

Rupabhinda, S. (2018). A survey of memory management techniques for virtual machines. *International Journal of Novel Research and Development*, 3(5), 46–48.

The Linux Kernel documentation. (n.d.). *zram: Compressed RAM-based block devices*. The Linux Kernel. <https://docs.kernel.org/admin-guide/blockdev/zram.html>